05. Positive-geometry tests for a fixed gravity integrand

Pure Thought research portfolio

Revised 17 September 2026

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

Particle-collision formulas often contain many terms whose cancellations hide a simpler structure. In some theories, a geometric shape organizes the answer: its boundaries encode the ways a collision can split into simpler events. This problem asks whether a tightly specified class of shapes can do the same for one gravity calculation. It treats geometry as a possible explanation for the structure of a formula, not as a literal picture of spacetime.

Key terms

- **Integrand** — The mathematical expression to be integrated to obtain a quantity such as a scattering amplitude.
- **Loop** — A feature of a quantum correction involving internal momenta that must be integrated over; one loop is the first such level here.
- **Supergravity** — A theory combining gravity with supersymmetry, a proposed relation between types of particles; N=8 specifies the chosen four-dimensional model.
- **Positive geometry** — A region with mathematical boundary properties that determine a special differential form.
- **Canonical form** — That distinguished differential form, whose singularities and boundary residues obey precise recursive rules.
- **Residue** — The coefficient describing the leading contribution of a simple pole; here it encodes boundary information.
- **Factorization** — The way a scattering process separates into simpler processes near certain physical singularities.
- **Ansatz** — A deliberately restricted class of candidate expressions or geometric constructions searched for a solution.
- **Pole at infinity** — A singularity revealed when a coordinate is taken to infinity, which must be included in a global comparison.

Why this matters

A geometric organization could explain cancellations and suggest more efficient calculations or new identities in quantum gravity. Even a carefully limited impossibility theorem is informative: it tells us which geometric proposals are too restrictive and where a different idea is needed.

What progress would mean

Success would mean either a verified shape-to-form correspondence or a proof that the declared candidate class cannot work. Failure to find a shape would not rule out every possible geometric

description of gravity.

Core question

For a specified low-loop supergravity integrand, does a precisely bounded class of candidate positive geometries reproduce its differential form, including every boundary and pole at infinity?

Scope and assumptions

Start with the four-point one-loop N=8 supergravity integrand, in a declared four-dimensional representation and helicity sector. Record any regulator and the limits of four-dimensional cut reconstruction. Derive or reproduce the target before searching. Tree-level and scalar-box canonical forms are calibration examples.

Define the search class before claiming an obstruction: ambient coordinates, dimension, allowed rational maps, inequalities, permitted external prefactors, and bounds on degree/number of boundaries. "No geometry exists" is not a finite computational question without such restrictions.

Mathematical target

Compare the top-form $\Omega_target=I(\ell;k)d^4\ell$ to $f(k)\pi_^*\Omega_G$ for a declared map π and external factor f. State whether equality is literal rational-form equality or equality modulo a specified class of terms integrating to zero. Work on a defined real slice with orientations.

For a standard positive geometry, Ω_G has logarithmic boundary poles and recursively normalized residues. Check all finite boundaries, their intersections, and infinity. Distinguish a canonical differential form from the symbol of an integrated polylogarithmic function.

Required outputs and proof obligations

Return either an explicit G, map, canonical form, and exact equality check, or a proof excluding the declared ansatz class. A failed search returns "unresolved within budget." A dlog decomposition alone is a candidate diagnostic, not a geometry certificate.

Validation and rejection controls

Verify simplex/box normalization independently. Check factorization and generalized cuts of the target. Rational reconstruction must be followed by an exact polynomial identity after denominators are cleared, with singular loci accounted for. Check orientation signs and residues at infinity explicitly.

Milestones and research extension

Baseline: exact target and pole/residue audit. Strong: one verified representation or one restricted no-go theorem. Research extension: broaden the geometry class or multiplicity; explain how the result changes understanding of gravity amplitudes rather than treating a computational failure as a universal obstruction.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- Arkani-Hamed, Bai and Lam, Positive Geometries and Canonical Forms — Definition of logarithmic canonical forms and boundary recursion.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

$$\Omega_{\text{target}} = I(\ell; k)\, d^4\ell, \qquad \Omega_{\text{target}} \stackrel{?}{=} f(k)\, \pi_* \Omega_G.$$

For a standard positive geometry, each oriented boundary must satisfy the canonical residue relation $\mathrm{Res}_{\partial G}\, \Omega_G = \Omega_{\partial G}$. Fix the geometry class, maps, prefactors, representation and equality convention first. Verify finite boundaries, intersections and infinity; an obstruction is limited to that class. An integrated symbol is not the integrand's canonical form.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Blocking — undefined search universe (35–45, 91–95).** An unrestricted geometry-existence question cannot be ruled out by an unsuccessful symbol search. The rewrite makes both positive and negative answers mathematically scoped.
2. **Major — form versus integral (53–63).** A symbol is not a wedge of dlog one-forms, and integrated branch-cut data are not interchangeable with poles of an integrand. Canonical-form residue recursion applies to a specified differential form.
3. **Major — kinematics (49).** All physical Mandelstam invariants need not be positive simultaneously. Positivity refers to the chosen real geometry and coordinates.
4. **Major — validation (84–89).** Agreement on selected cuts or numerical integration can miss contact terms, infinity poles, or representation ambiguities. The new deliverable requires an exact equality statement and an exhaustive boundary audit within the chosen ansatz.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a specified low-loop supergravity integrand, does a precisely bounded class of candidate positive geometries reproduce its differential form, including every boundary and pole at infinity?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: exact target and pole/residue audit. Strong: one verified representation or one restricted no-go theorem. Research extension: broaden the geometry class or multiplicity; explain how the result changes understanding of gravity amplitudes rather than treating a computational failure as a universal obstruction.

Sources supporting the corrections

- Arkani-Hamed, Bai and Lam, Positive Geometries and Canonical Forms — Definition of logarithmic canonical forms and boundary recursion.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.

- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.